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Access level Answers
by UnMonsieur

How will you detect proxy cheats

Proxy clients often are meant to work on specific servers (for example, Hypixel), through a host that in some way connects to the target server with a different IP. Most of the proxy clients have died out, latest largest proxy that I am aware of is [Lilith](#)

Let's review how Lilith works

```
Lilith Launcher Stable Release 3
=====
Lilith v1.0.35 [...] (node:26772) Warning: Setting the NODE_TLS_REJECT_UNAUTHORIZED environment variable to '0' makes TLS
connections and HTTPS requests insecure by disabling certificate verification.
(Use `lilith-win-1-0-35 --trace-warnings ...` to show where the warning was created)
Lilith v1.0.35 [bedwarstourney#0]
Config » Fetching... (node:7824) Warning: Setting the NODE_TLS_REJECT_UNAUTHORIZED environment variable to '0' makes TLS
connections and HTTPS requests insecure by disabling certificate verification.
(Use `lilith-win-1-0-35 --trace-warnings ...` to show where the warning was created)
Config » Fetched! Change at https://me.lilithmod.xyz!
(node:30512) Warning: Setting the NODE_TLS_REJECT_UNAUTHORIZED environment variable to '0' makes TLS connections and HT
PS requests insecure by disabling certificate verification.
(Use `lilith-win-1-0-35 --trace-warnings ...` to show where the warning was created)
Ready » Join Lilith using the address below:

Server Address
localhost
```

Proxy clients are *in theory* meant to improve the user's PING for smoother connection. Although there are private clients that are meant for cheating. Here's a prime example from one of Headed's old [Videos](#)

Detecting Proxy clients requires the players to remain on the server while they're being ScreenShared. Here are some ways and theories on how unreleased / private proxy clients can potentially be detected. *Let's start with unethical ways.* . .

Requiem has released a video detecting Proxy Clients by IP & Geolocation monitoring. While RedLotus does not approve of this specific detection, your server(s) might, considering you're a trusted ScreenSharer and not an idiot.

<https://www.youtube.com/watch?v=nHLFN0trMTk>

An alternative to this would be using System Informer or TCPView by Sysinternals for a more ethical and secure approach, hence we will be analyzing the **remote address** javaw.exe is receiving packets from.

In System Informer, go to **Network** Processes Services Network
TCPView will automatically display Network activity.

System Informer

Name	PID	Local address	Local...	Remote address	Remote port	Prot...	State	Owner
javaw.exe	13860		53479	209.222.114.10	25565	TCP	Established	
javaw.exe	13860		53532	138.199.37.231	443	TCP	Established	
javaw.exe	13860		53488	138.199.37.231	443	TCP	Established	
javaw.exe	13860		53441		443	TCP	Established	
javaw.exe	13860		53537		80	TCP	Established	
javaw.exe	13860		53530		80	TCP	Established	
javaw.exe	13860		53537					
javaw.exe	13860		53532					
javaw.exe	13860		53530					
javaw.exe	13860		53488					
javaw.exe	13860		53479					
javaw.exe	13860		53441					



To accurately determine if the user is on a proxy client or not, you need to know the Remote Address of your server in the first place, so use this to gather it from your device first.

Analyzing chat logs for server IP connections and the IP of what you've found is the next step after using TCPView and SystemInformer. While Wireshark and Geolocation analysis is definite, this is ethical and does not put players in danger of malicious staff members.

Theorising

One theory for catching private proxy clients is that they need a definite host to connect to, which means the user will most definitely be running a script or a terminal / command prompt in the background to transfer the connection towards Minecraft. Looking for suspicious scripts, .exes or terminals may work.

Chat logs and servers list can help with detecting suspicious connections.

Event Logs bypass

First of all, unless the Eventlog service is disabled - these logs can not be bypassed or overwritten.

However, event log tracking can be disabled without the need to disable the service itself entirely. One of the easiest ways is to utilize PsExec to “hijack” a specific event log from the event viewer by disconnecting it. [> DEMO <](#)

Obviously enough, detecting this is also not an issue considering we will still be able to see the latest entry. If they clear event logs, that can also be traced by event ID 1102. Finding PsExec itself on the current system / current boot is enough to ban for.

Java bytecode execution traces

Detection to this is not mentioned but still possible with the Kernel live dump method covered in the free Red Lotus guide. However, in most scenarios a bytecode execution will leave a .class file behind, using USN Journal by “File Create” > “.class” can guide you thru this. Otherwise using the methods by prefetch mentioned in the Red Lotus server will do most of the trick because it will log what file was executed, what file was dropped and so on.

When it comes to .exes, they may drop a .jar that they use for such execution.

Theorising

While analyzing this behavior, extracting \$MFT and \$UsnJrnl:J is a good idea. A collection of Prefetch data (WinPrefetch or PECmd) with \$MFT and \$J, as well as Kernel Live Dump can help with analyzing what was the following action after executing CMD, Powershell, or any Terminal / Scripting environment they have (Vscode, nodejs, python etc)

Use the dates collected from suspicious files / CMD executions from MFT to analyze the USN Journal and Kernel Dump. The rest is on you.

Cheats that don't load anything with AnyDesk

First things first - we need to even assume what we've found is a cheat. Most cheats are unsigned and most cheats require Administrator, thus they have a shield icon on. Most legitimate windows files - don't. If nothing happens when you run them, try running them from CMD (cd to the directory (cd path) and run the .exe)

Other ways of detecting this is by simply asking the Player to share their screen through Discord whilst having AnyDesk closed or ask them to send you the executable (keep in mind that you can't ban if they don't unless you have a strong reason to believe it is a cheat) a strong reason would be:

- If you upload the file on virustotal and it returns as a virus, its behavior is strange and abnormal;
- If the file has no signature + its zone.identifier is from a fishy web (or cdn.discordapp) (in that case you can download the file itself on your device as well)
- If the file has a shield icon and the user has in fact executed it himself in current system boot.
- You analyze the file with DiE (detect it easy) and find relations to AnyDesk

Renamed extension with no traces

System Informer can't be bypassed unless the file's extension includes a Chinese character since it can't read the Unicode of it. However, it leaves a trace with question marks. Simply including a question mark and a dot in the filter of whatever service you need (so ? > . > :\) or using the DPS method for catching renamed extensions (refer to SS Community > Regex) will do.

If the file is an executable, it will leave a trace in prefetch, depending on its execution way. Such files are often executed using the WMIC bypass method.

In the end, using a search everything regex for unicode characters (refer to SS Community > Regex) will do the trick.

For files with no extension, keep in mind that these files can NOT be run even through CMD, except for the WMIC, so filtering search everything for extension-less files (ext:) (space after ext:) and sorting by size (or limiting the size) may work. So:

size:3mb..15mb ext:

And frankly, in theory these types of files should leave a trace in jumplists and shell:recent, so... (or USN Journal > renamed from & renamed to > ?) question marks will do 😊

Kernel live dump

For ScreenSharing purposes, Kernel Live dump is a mini-dump of the memory in current system boot, current instance. It can show file executions, CMD commands, powershell commands, file downloads and so on. Its main use would be to detect process hollowing, regsvr / rundll injections and so on. (All covered in the free guide)

Startup XML

Instead of using task scheduler, some users utilize the startup functionality of Windows to run cheats on system boot. Startup data is logged in

C:\Windows\System32\WDI\LogFiles\StartupInfo

Every startup .xml starts with a S-

S-1	
Name	Path
 S-1-5-21-3437053527-345174875-1567029899-1...	C:\Windows\System32\WDI\LogFiles\StartupInfo

The directory is inaccessible by default, but just double clicking on the .xml file opens what we need.

```
<?xml version="1.0" encoding="UTF-16"?>
<StartupData IntervalStartMs="9177" IntervalEndMs="99177">
  <Process Name="C:\Windows\System32\SecurityHealthSystray.exe" PID="7628" StartedInTraceSec="20.198">
    <StartTime>2023/09/21:22:39:57.9196292</StartTime>
    <CommandLine><![CDATA["C:\Windows\System32\SecurityHealthSystray.exe" ]]></CommandLine>
    <DiskUsage Units="bytes">10240</DiskUsage>
    <CpuUsage Units="us">40441</CpuUsage>
    <ParentPID>6768</ParentPID>
    <ParentStartTime>2023/09/21:22:39:47.1723716</ParentStartTime>
    <ParentName>explorer.exe</ParentName>
```